SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I ________K .Deeksha Sree(192424097)___________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K .Deeksha Sree

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart Home Automation System Using Intelligent Agents
Introduction
A Smart Home Automation System uses Artificial Intelligence (AI) to automatically control lighting, temperature, and security based on user behavior and environmental conditions. It receives real-time inputs from sensors such as motion detectors, temperature sensors, and clocks, while learning user preferences to improve comfort, energy efficiency, and security.

Types of Intelligent Agents and Their Functions
1. Simple Reflex Agent
Definition:
A Simple Reflex Agent makes decisions based only on the current input using predefined IF–THEN rules. It does not remember past events.
Working in Smart Home:
- If motion is detected → Turn ON the light.
- If no motion is detected → Turn OFF the light.
- If temperature exceeds 30°C → Turn ON the air conditioner.

Advantages
- Fast response.
- Easy to implement.
- Suitable for simple tasks.

Disadvantages
- Cannot learn user preferences.
- Cannot handle complex situations.
- No memory of previous states.

2. Model-Based Reflex Agent
Definition:
A Model-Based Agent maintains an internal model of the environment and remembers previous states before making decisions.
Working in Smart Home
- Remembers whether a person recently entered a room.
- If motion temporarily stops while someone is sleeping, it keeps the lights OFF instead of assuming the room is empty.
- Tracks temperature changes over time before adjusting cooling.

Advantages
- Better decision-making than Simple Reflex Agent.
- Handles partially observable environments.
- Uses past information.

Disadvantages
- More complex.
- Requires additional memory and computation.

3. Goal-Based Agent
Definition:
A Goal-Based Agent selects actions that help achieve specific goals.
Working in Smart Home
Possible goals:
- Maintain room temperature at **24°C**
- Reduce electricity consumption

- Keep the home secure

Example:
If electricity usage is high, the agent turns OFF unnecessary appliances while ensuring user comfort.

Advantages

- Plans actions intelligently.
- Can compare different possible actions.
- More flexible.

Disadvantages

- Slower than reflex agents.
- Requires planning algorithms.

4. Utility-Based Agent

Definition:
A Utility-Based Agent selects the action that provides the highest overall benefit (utility) by balancing multiple objectives.

Working in Smart Home

The agent considers:

- User comfort
- Energy efficiency
- Security
- Cost

Example:
Instead of setting AC to 18°C immediately, it chooses 24°C because it provides sufficient comfort while saving electricity.
Similarly,

- Brightness is adjusted according to daylight.
- Security cameras become more active at night.
- Heating/Cooling is optimized to minimize energy costs.

Advantages

- Produces optimal decisions.
- Balances conflicting objectives.
- Learns user preferences over time.

Disadvantages

- Computationally expensive.
- Utility function must be carefully designed.

Comparison of Intelligent Agents

Agent Type	Memory	Learning	Goal-Oriented	Best Use
Simple Reflex	No	No	No	Immediate responses
Model-Based	Yes	Limited	No	Uses current and past states
Goal-Based	Yes	Possible	Yes	Achieves defined objectives
Utility-Based	Yes	Yes	Yes	Maximizes comfort, security, and efficiency

Most Effective Agent (Justification)

A **Hybrid Agent** combining **Model-Based**, **Goal-Based**, and **Utility-Based** approaches is the most effective solution.

Why Hybrid?

Model-Based Component

- Maintains information about occupancy.
- Remembers previous temperature and lighting conditions.
- Handles incomplete sensor information.

Goal-Based Component

- Achieves predefined goals such as:
    - Comfortable temperature
    - Home security
    - Energy saving

Utility-Based Component

- Chooses the best action among several alternatives.
- Balances comfort, cost, and safety.
- Learns user routines and preferences over time.

Example Scenario

Situation:

- Time = 10 PM
- User enters living room.
- Outside temperature = 34°C.
- Electricity tariff is high.

Hybrid Agent Decision

- Turns ON only required lights.
- Sets AC to 24°C instead of 18°C.
- Closes smart curtains.
- Activates outdoor security cameras.
- Switches OFF unused appliances.

This provides:

- Comfort
- Energy efficiency
- Security

Conclusion

Although Simple Reflex and Model-Based Agents are suitable for basic automation, they cannot optimize multiple objectives. A Hybrid Agent combining Model-Based, Goal-Based, and Utility-Based intelligence provides the best balance between comfort, security, and energy efficiency while adapting to user behavior over time.

2.

Robot Navigation in an Unknown Maze Using Uninformed Search

Introduction

A robot placed in an unknown maze must discover a path to the exit without prior knowledge of the environment. Since no heuristic information is available, uninformed search algorithms such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) are suitable approaches.

Uniform Cost Search (UCS)

Definition

Uniform Cost Search expands the node with the lowest cumulative path cost first.
It guarantees finding the least-cost path.

Working in the Maze
Suppose different paths have different movement costs.
Example:

- Straight path = Cost 1
- Rough terrain = Cost 5
- Slippery floor = Cost 3

The robot always selects the path with the smallest total cost until it reaches the exit.

Advantages

- Complete.
- Optimal solution.
- Finds the minimum-cost path.
- Suitable when movement costs vary.

Disadvantages

- High memory consumption.
- Slow for large search spaces.
- Expands many nodes.

Complexity

Measure	UCS
Time Complexity	$O(b^{(C^*/\varepsilon)})$
Space Complexity	$O(b^{(C^*/\varepsilon)})$

where:

- b = branching factor
- C* = optimal solution cost
- ε = minimum step cost

Iterative Deepening Search (IDS)
Definition
Iterative Deepening Search repeatedly performs Depth-Limited Search with increasing depth limits until the goal is found.

Working in the Maze
Search process:
Depth = 0
Robot checks only the starting node.
Depth = 1
Robot explores immediate neighbors.
Depth = 2
Robot explores two levels deep.
This continues until the exit is discovered.

Advantages

- Complete.
- Uses very little memory.

- Suitable for large search spaces.
- Finds shallow solutions quickly.

Disadvantages
- Repeats node expansions.
- Not optimal if movement costs differ.
- Extra computation due to repeated searches.

Complexity

Measure	IDS
Time Complexity	O(b^d)
Space Complexity	O(bd)

where:
- b = branching factor
- d = depth of the shallowest goal

Comparison of UCS and IDS

Feature	UCS	IDS
Path Cost	Minimum Cost	Minimum Depth
Complete	Yes	Yes
Optimal	Yes	Only with equal step costs
Time	High	Moderate
Space	Very High	Very Low
Best For	Different path costs	Unknown depth with limited memory

Relation to Intelligent Agents
Simple Reflex Agent
- Chooses actions only from current sensor input.
- May repeatedly hit dead ends.
- Cannot remember explored paths.

Not suitable.

Model-Based Agent
- Maintains a map of explored areas.
- Avoids revisiting previous locations.
- Learns maze structure while exploring.

Suitable.

Goal-Based Agent
Goal:
Reach the maze exit.
The robot evaluates possible paths that move it closer to the goal.
Highly suitable.

Utility-Based Agent
The robot evaluates:
- Travel distance
- Energy consumption

- Time taken
- Safety

It selects the path with maximum overall benefit.
Most intelligent.

Effect of Agent Design on Decision-Making
Agent design directly affects navigation performance.

- Simple Reflex Agent reacts immediately but cannot learn.
- Model-Based Agent remembers explored paths, reducing repeated exploration.
- Goal-Based Agent focuses on reaching the exit efficiently.
- Utility-Based Agent considers multiple factors such as time, energy, and safety before selecting the best path.

Better agent designs lead to faster, safer, and more efficient navigation.

Most Effective Combination (Justification)
The most effective solution is a Hybrid Agent (Model-Based + Goal-Based + Utility-Based) using Uniform Cost Search (UCS).
Justification

- Model-Based maintains a map of the maze and avoids revisiting explored areas.
- Goal-Based keeps the robot focused on reaching the exit.
- Utility-Based optimizes decisions by considering path cost, energy consumption, and safety.
- Uniform Cost Search guarantees the least-cost path when movement costs vary, making it ideal for real-world mazes with different terrains.

If all movements have equal cost and memory is limited, Iterative Deepening Search (IDS) is a practical alternative because it requires significantly less memory while still guaranteeing that a solution will be found.

Conclusion
Uniform Cost Search is the preferred search strategy when path costs differ because it guarantees the optimal path, whereas Iterative Deepening Search is suitable for memory-constrained environments with equal step costs. A Hybrid Intelligent Agent combining Model-Based, Goal-Based, and Utility-Based capabilities with Uniform Cost Search provides the best overall performance by enabling efficient navigation, minimizing travel cost, conserving energy, and making intelligent decisions in an unknown maze.